

Polyline Editor – System Analysis

1. Scope Definition

The scope of this project is to design a simple and interactive Polyline Editor that allows users to create, edit, and manage polylines efficiently using keyboard and mouse input. The system focuses on drawing polylines, editing points (move/delete), managing multiple polylines, and providing real-time feedback.

The system is designed for students and beginner-level users, aiming to reduce complexity found in professional tools like CAD software.

2. Functional Requirements

- Begin a new polyline using key 'b'
- Add points by clicking on the canvas
- Move points using key 'm' and dragging
- Delete points using key 'd'
- Refresh the canvas using key 'r'
- Exit the program using key 'q'

3. Non-Functional Requirements

- Provide instant visual feedback
- Be simple and easy to use
- Maintain consistent interaction behavior
- Support error prevention and recovery
- Be responsive
- Ensure clear interface and status messages

4. HCI Principles / Guidelines

Feedback:

The system provides immediate and clear feedback after every user action. When a user adds, moves, or deletes a point, the change is instantly visible on the canvas. Selected points are highlighted, and status messages inform the user about the current action. This helps users understand what is happening and reduces confusion.

Direct Manipulation

The system supports direct manipulation by allowing users to interact with graphical elements directly. Users can click to create points and drag them to new positions. This makes the interaction more natural, intuitive, and closer to real-world behavior, improving usability.

Consistency

The interface maintains consistency by ensuring that the same keys always perform the same actions (e.g., 'm' for move, 'd' for delete). Interaction patterns remain uniform throughout the system, which helps users learn quickly and reduces the need for repeated learning.

Affordance

The design provides clear visual cues that indicate how elements can be used. Points appear selectable and draggable, and shapes are clearly defined. These visual affordances guide users on what actions are possible without needing instructions.

Visibility of System Status

The system clearly displays the current mode (e.g., draw, move, delete) and provides status messages after each action. This ensures that users are always aware of what the system is doing and what mode they are in, reducing ambiguity.

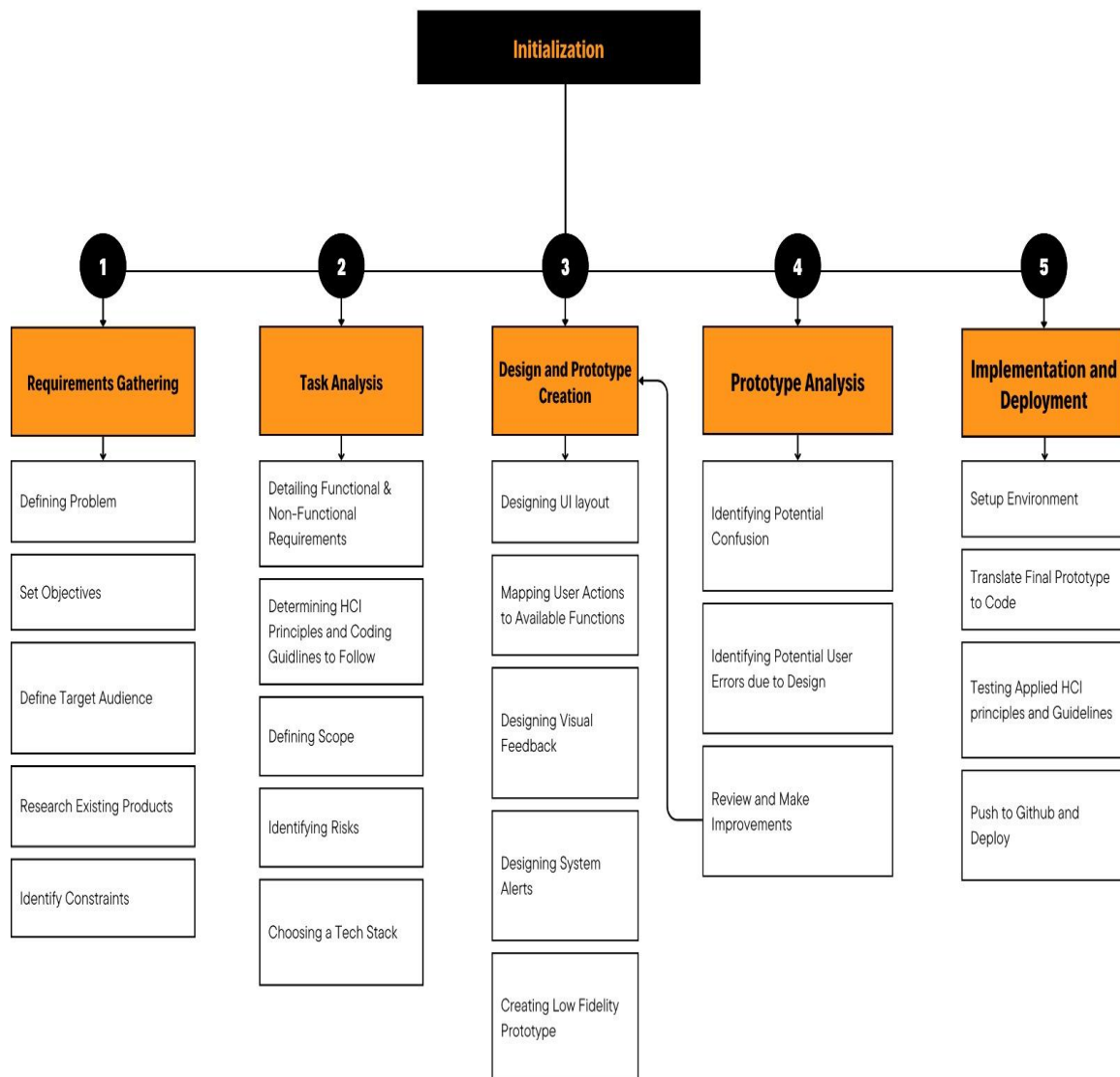
5. Risks and Limitations

- Incorrect point selection
- Accidental deletion
- Visual clutter with many polylines
- Shape complexity

6 .User-Centered Workflow for Polyline Editor

Polyline Editor

HCI and Computer Graphics Mid - Term



7 .HIERARCHICAL TASK ANALYSIS OF POLYLINE EDITOR

